Structured summary: | Power Analysis for Run-In Clinical Trials under AR(1) and General Correlation Structures

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This document is a structured, section-by-section summary of the key points argued in the companion manuscript, "| Power Analysis for Run-In Clinical Trials under AR(1) and General Correlation Structures." It is provided as a supplementary reading aid and does not stand on its own; all notation, derivations, simulation detail, and citations are in the main manuscript.

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1 Introduction

The efficiency of run-in designs is governed by the correlation structure of the repeated measurements.

- Under compound symmetry the correlation between change scores is constant.
- Run-in observations then inform the random slope through simple averaging.
- Under AR(1) the correlation decays with separation, so run-in information depends on distance from the treatment period.

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This paper extends prior compound-symmetric run-in frameworks to AR(1) residual correlation.

- Paper 1 assumed the rank-one-plus-identity change-score covariance from independent errors.
- Wang (2019) extended run-in to two-period mixed models but kept compound symmetry.
- We instead adopt AR(1) residual correlation with geometric decay in lag.

Existing literature treats autocorrelation and change-score variation but not the run-in setting.

- Winkens (2007) studied optimal designs for autocorrelated data outside the run-in context.
- Diggle and Verbeke give general treatments of longitudinal correlation structures.
- Munoz et al. (1992) embedded compound symmetry and AR(1) in a single damped-exponential family, the natural generalization of the two structures contrasted here.
- Chambless (1993) addressed intra-individual variation in change scores, a concern amplified under AR(1).

The closed-form AR(1) derivation covers a design case the reference longpower toolkit does not.

- Its slope-based closed-form routines assume independent or compound-symmetric residuals.
- Its `power.mmrm.ar1()` routine implements the analytic AR(1) formula of Lu, Luo, and Chen (2008), but without run-in or common close observations under a random slope.
- Our formula supplies that case and collapses to the compound-symmetric structure of Paper 1 as $\rho \to 0$.

1.1 Marginal covariance

The AR(1) change-score covariance loses rank-one structure but the Woodbury reduction survives.

- The matrix R is no longer rank-one-plus-identity, so Sherman-Morrison does not apply.
- The Woodbury identity still reduces the inverse covariance to a scalar inversion.
- This holds because the random-effects covariance G remains scalar.

1.2 Woodbury computation

The Woodbury scalar term persists, but the residual inverse must now be computed differently.

- The correction term H remains a scalar as in Paper 1.
- The residual inverse R^{-1} must be obtained numerically.
- Equivalently it follows from the AR(1) tridiagonal inverse rather than Sherman-Morrison.

1.3 Cross-covariance between phase means

A moment-based alternative derives the pre-post variance directly from the phase means.

- It computes the variance of the test statistic from the moments of the phase means.
- The run-in mean averages the run-in observations.
- The treatment-period mean averages the treatment-period observations.
- The construction follows the summary-measures approach of Frison and Pocock (1992).

2 Relationship between the two approaches

The GLS and moment-based approaches target distinct variance quantities.

- The GLS variance is the minimum-variance bound under the assumed model.
- The moment-based variance is that of the simple averaged change-score estimator.
- The two therefore answer different efficiency questions.

The two approaches do not coincide even without a random slope.

- At $\sigma_b^2 = 0$ the GLS estimator still weights observations through the AR(1) residual covariance and targets the slope difference.
- The moment-based quantity is the variance of the unweighted phase-mean difference, a different estimand.
- Numerically, at $J_0 = 2$, $J_1 = 2$, $\rho = 0.5$, $\sigma^2 = 1$, $\sigma_b^2 = 0$: GLS 0.636 versus moment-based 1.219.

The moment-based computation serves the simple pre-post estimator, not as a bound on the GLS variance.

- It requires no specification of the random-slope variance.
- It is the exact variance of the unweighted pre-post difference an analyst might report without fitting a mixed model.
- Because the estimands differ, its ratio to the GLS variance is descriptive, not an efficiency bound.

3 Discussion

The autocorrelation structure matters for run-in designs only when the autocorrelation is weak relative to the random-slope signal.

- When the slope-to-noise ratio is large, the random-slope term dominates the change-score covariance and the residual correlation structure is second order.
- The AR(1) and compound-symmetry variances then differ negligibly, and the simpler Paper 1 formula suffices.

- The two structures diverge most when the slope-to-noise ratio is small and run-in observations are distant from randomization.

At a matched nominal ρ, compound symmetry is generally conservative, with a narrow anticonservative corner.

- Positive serial correlation reduces the residual variance of every change score, so the AR(1) variance falls below the compound-symmetric value for most designs.
- The discrepancy grows with ρ; near $\rho = 0.95$ the AR(1) variance can fall below one tenth of the compound-symmetric value.
- The exception is large J_0 at small ρ, where distant run-in observations lose information and compound symmetry understates the variance by at most about 6%.

The numerical comparisons give a practical decision rule for when the simpler formula is safe.

- At a matched nominal ρ, the compound-symmetry formula of Paper 1 is a conservative and convenient default for most designs.
- With many run-in observations and low expected autocorrelation, the AR(1) formula should be used directly; the compound-symmetric formula can understate the variance by up to about 6% there.
- When ρ is uncertain, evaluating the AR(1) formula over a plausible range of ρ and taking the largest variance is a safe default.

4 Data availability statement

The implementing package, tests, and bibliography are openly available in the project repository.

- The package implements the AR(1) variance formulas.
- Validation tests and the shared four-paper bibliography are included.
- Specific paths appear in the Reproducibility section below.

5 Reproducibility

The manuscript was rendered with R 4.4.x using a small set of principal packages.

- The in-package source supplies the AR(1) machinery.
- Testing uses the tinytest harness.
- The manuscript build uses rmarkdown and bookdown.

All reported quantities are deterministic.

- Given fixed inputs the variance formulas exercise no random-number generator.
- No Monte Carlo computation appears in this manuscript.
- Re-running the chunks therefore reproduces every table and figure exactly.

All source, companion manuscripts, and bibliography reside at documented repository paths.

- The R package source is in the package source directory.
- The companion manuscripts for Papers 1, 3, and 4 have their own analysis subdirectories.
- This manuscript source and the shared bibliography reside in the correlation-paper directory.